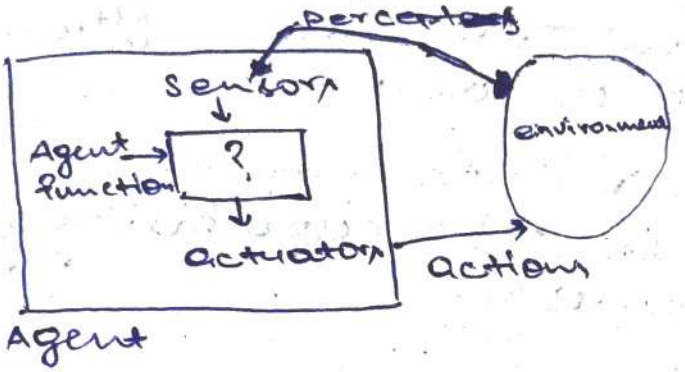
Artificial Intelligence

- Ability to acquire & apply knowledge equally.
- ✓ capacity for Logic
- ✓ Understanding
- ✓ Self-awareness
- ✓ learning
- ✓ Emotional knowledge
- ✓ Reasoning
- ✓ Planning
- ✓ Creativity
- ✓ Critical thinking & problem solving



- It is a program that acts as humans do.

① Thinking Humanly

- The automation of activities that we associate with human thinking, problem solving & learning.
  [Bellman, 1978]

② Thinking Rationally

- The study of mental faculties through the use of computational model.
  [Charniak & McDermott, 1985]
- Study of computations that make it possible to perceive reason & get.

③ Acting Humanly

- The art of creating machines that perform functions that require intelligence when performed by people.
  [Rich & Knight, 1991].

(4) Acting Rationally

- Computational intelligence is the study of the design of intelligent agents. [POOLE et al, 1998]

- Agent = Architecture + Agent function

Evaluation of A.I :-
↳ 3 stages

(1.) Artificial NARROW Intelligence (WEAK AI)

(2.) Artificial General Intelligence (Strong AI)
↳ Agent (=> Human Brain

(3.) Artificial Super Intelligence
↳ The time when the capability of computers will surpass humans.

Imp

Performance measure of any rational agents are done using 4 parameters —

"PEAS"
- P => Performance
- E => Environment
- A => Actuators
- S => Sensors

Ex.

	Performance	Environment	Actuators	Sensors
Taxi	Safety, time	Road, traffic, pedestrians, customers	Steering, brake, signal horn, accelerator	camera, sonar, GPS, speedometer, accelerometer
Chess Board.				

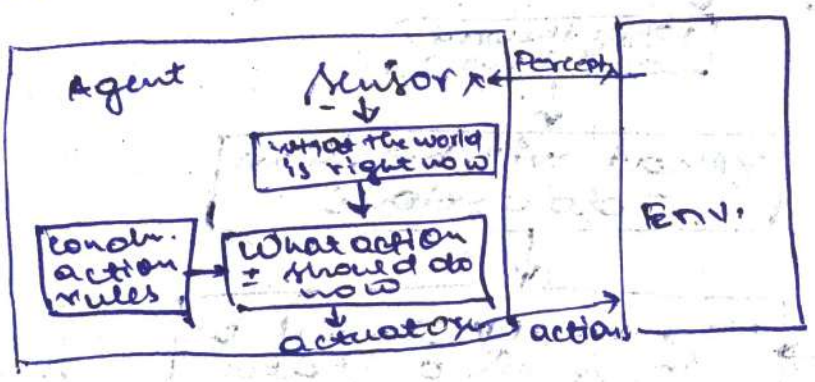
Types of Environment :-

1. Fully observable v/s Partial observable
2. Single Agent v/s Multi Agent
3. Deterministic v/s Stochastic
4. Episodic v/s Sequential
5. Static v/s Dynamic
6. Discrete v/s continuous
7. Known v/s Unknown

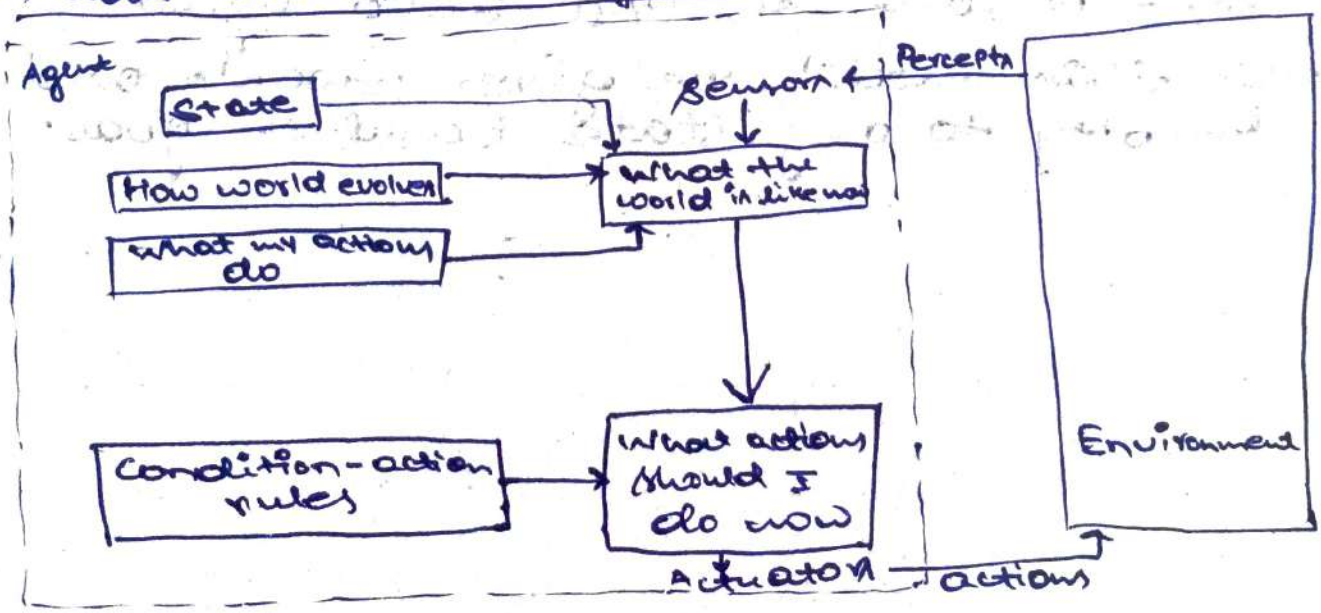
Types of Agents :-

1. Simple Reflex Agent

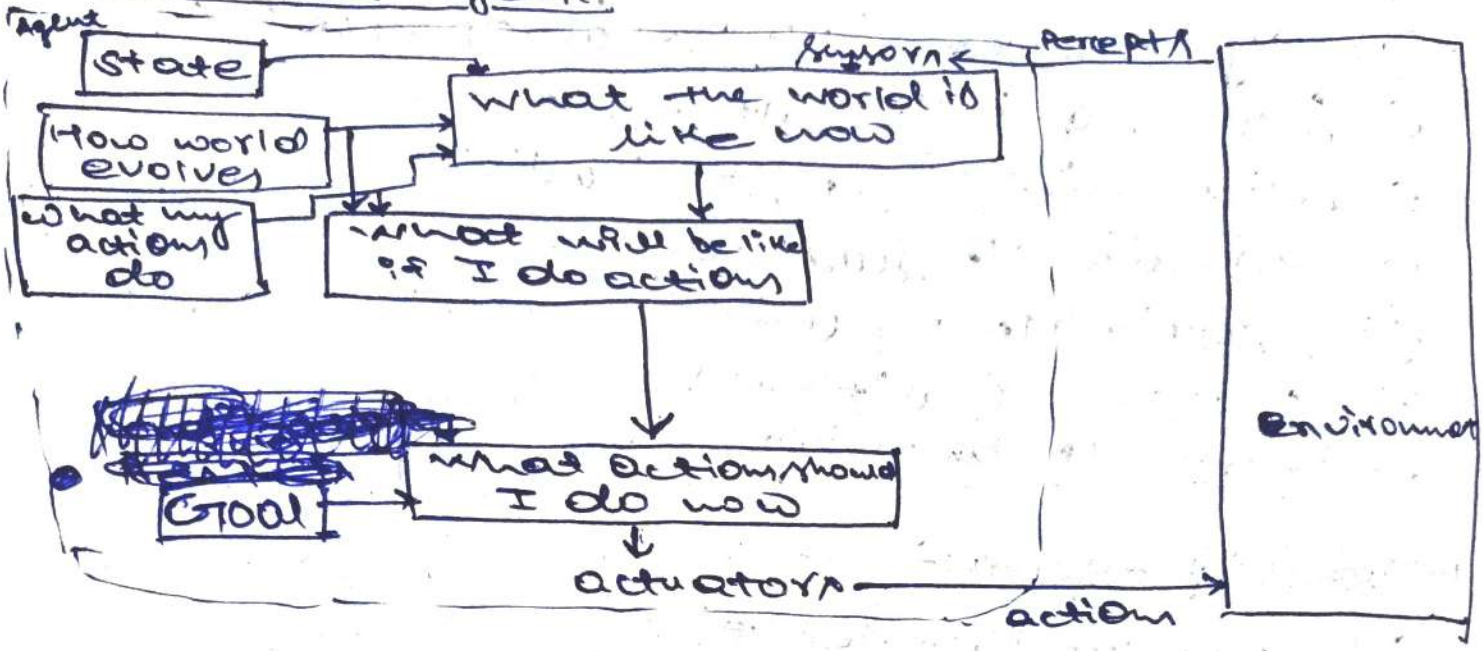
→ Based on the current state of the environment it performs the action. It does not care about the prior knowledge.



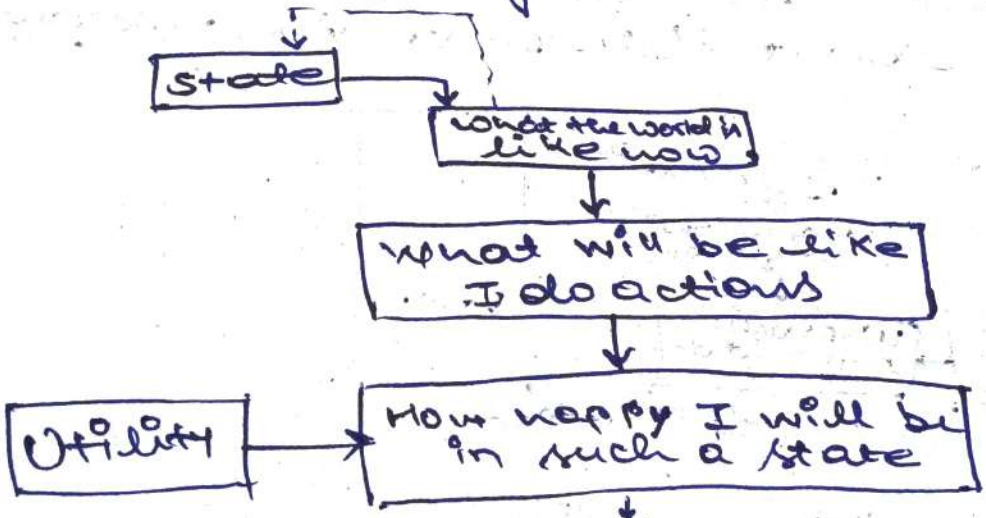
2. Model Based Reflex Agent



③ Goal Based Agent.

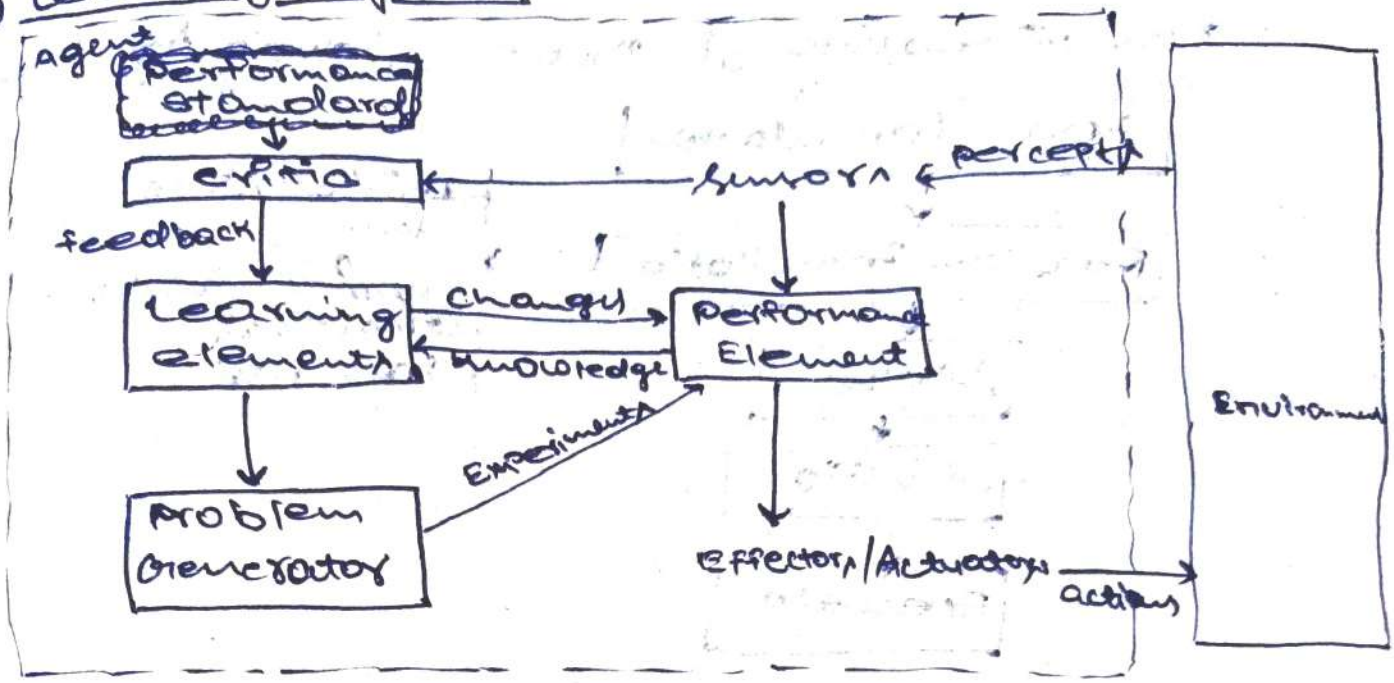


④ Utility Based Agent



- Rest is same as Goal Based agents.
- In this model, we give rewards and Penalty to any state based on goal.

⑤ Learning Agents



- Learning elements are responsible for making improvements by learning.

⑥ Multi Agent Systems

⑦ Hierarchical Agent

PSA → By Searching

Goal Based Agent

Env.
- → Fully Observable
- → Deterministic
- → Static
- → Known

Actions = {L, R, U, D}

S: {Start, Goal}

Ex.

1	2	3
5	8	4
7	0	6

3*3 →

1	4	7
2	5	8
3	6	

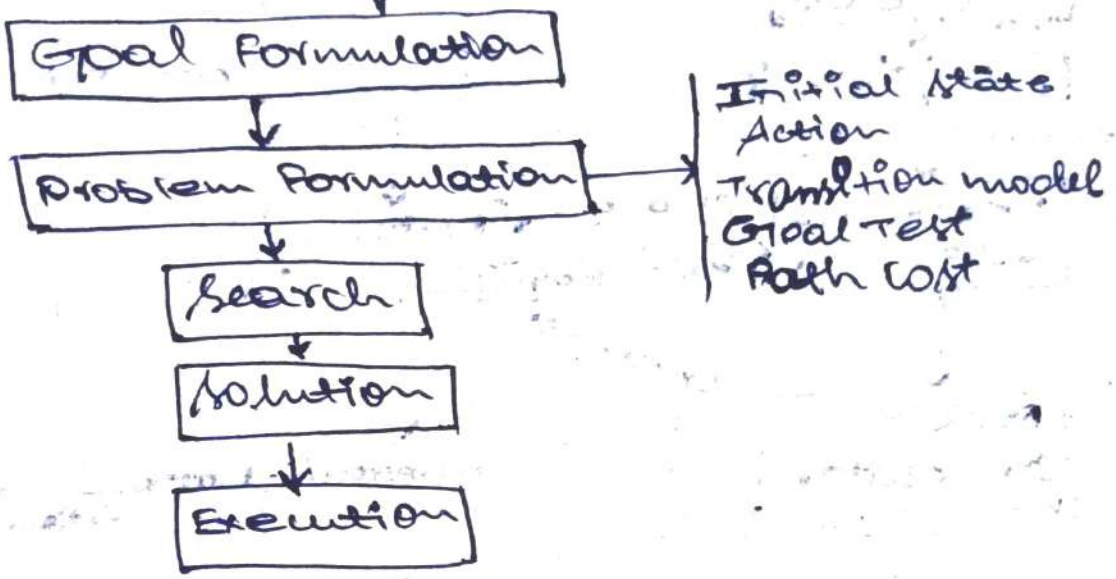
"State Space Search"

{S: {S, A, Action(S), Result(s,a), Cost(s,a)}

(In which state you are searching after applying action)

Result when you are reached after applying action on state S

→ Functionality of Problem Solving Agent:



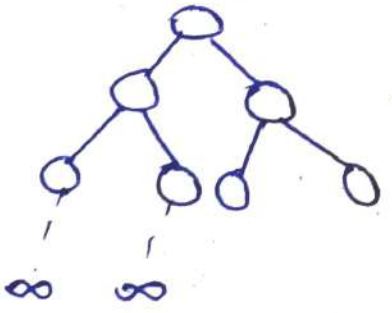
Problem Solving by Search :-

- well defined problem → Goal formulation
  Problem formulation
  searching by technique

ill-defined problem

1. Completeness
2. Optimality
3. Complexity w.r.t time
4. Complexity w.r.t space

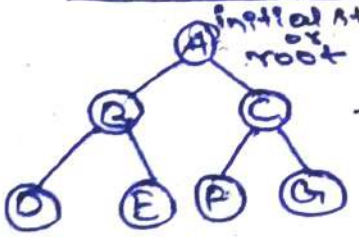
Ex.



→ (Problem not complete)

- Uninformed search → (DFS, BFS, DFID, ...)
  (blind search)
- Informed search
  ↳ Extra info. available for reaching the goal state

→ BFS (Breadth-First Search)



Visited node	Queue
	~~A~~
A	~~B~~ C
B	~~C~~ D E
C	~~D~~ E F G
D	~~E~~ F G
E	~~F~~ G
F	~~G~~
G	

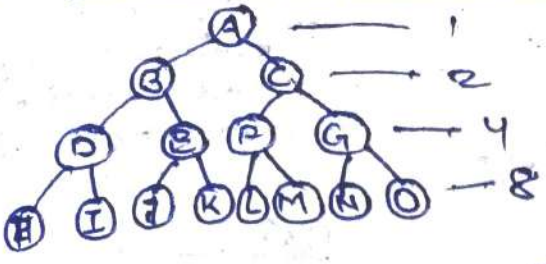
- Memory required : b^d ✩✩
  $O(b^d)$ (branching factor)

Complete ✓
Optimality ✓

Time required = $2^0 + 2^1 + 2^2 + \dots + 2^d$

$= \frac{2^0(2^{d+1}-1)}{2-1}$ | $\frac{b^{d+1}-1}{b-1} \rightarrow O(b^d)$ ✩✩

→ DFS (Depth First Search)





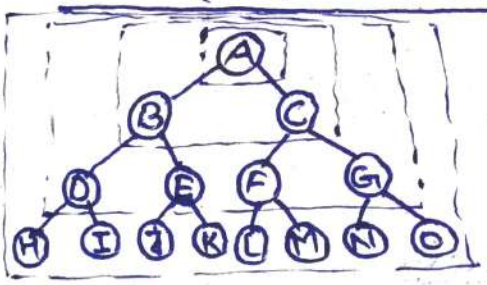
Visited nodes: A B D H I E J K C F L M G N O

- Memory Required : $(b-1)d+1$

$O(bd)$

- Time Required $= \frac{b^{d+1}-1}{b-1}$ → $O(b^d)$

→ DFID (Depth First Iterative Deepening)



Iter 0 A
Iter 1 A B C
Iter 2 A B D E F G
Iter 3 ~~A B C D E F G H I J K L M N O~~
A B D H I E J K C F L M G N O

- Memory Required : $(b-1)d+1$

Extra work done : $\frac{x-y}{x} * 100\%$

- Node visited:

No. of

$= 2^0 + (2^0+2^1) + (2^0+2^1+2^2) + (2^0+2^1+2^2+2^3) + \dots + (2^0+2^1+2^2+\dots+2^d)$

$= b^0 + (b^0+b^1) + (b^0+b^1+b^2) + \dots + (b^0+b^1+b^2+\dots+b^d)$

- $\sum_{l=0}^{d} \left(\sum_{i=0}^{l} b^i \right)$

DFS : $4^0+4^1+4^2+4^3+4^4 = 341$

BFS : $\frac{341}{4^0+4^1+4^2+4^3+4^4}$

No. $\frac{453-341}{453} * 100$

$= 24.72\%$

Ex: $b=4$, $d=4$

No. of nodes visited in DFID: $\sum_{l=0}^{4} \left(\sum_{i=0}^{l} 4^i \right)$

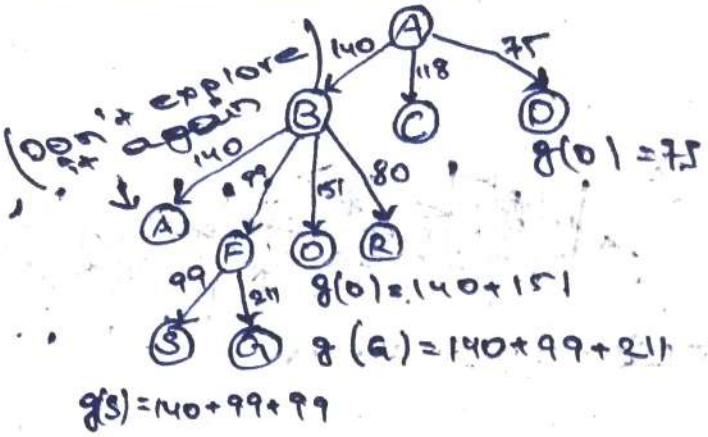
$= 4^0 + (4^0+4^1) + (4^0+4^1+4^2) + (4^0+4^1+4^2+4^3) + (4^0+4^1+4^2+4^3+4^4)$

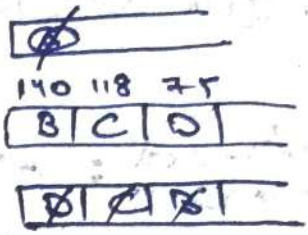
$= 1+5+21+85+341 = 453$

→ Uniform Cost Search → Priority Queue → $f(n) = g(n)$

$g(n)$ ↓ actual cost of node n from root node (A)

Ex.





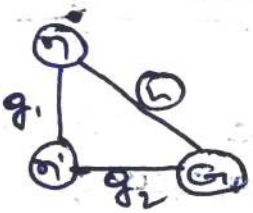
Order: A D C B R F O S G

Note:- A.I.A.O check bidirectional search

→ Heuristic Search (Informed)

- → Admissible → $h(n) \leq g^*(n)$
- → Non-admissible → Otherwise

Ex.



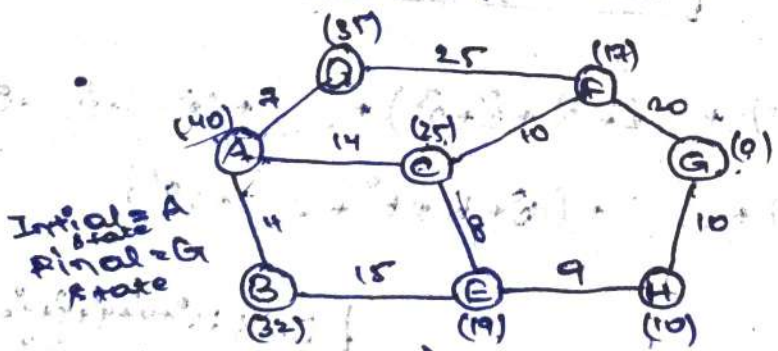
$h < (g_1 + g_2)$

J.V.I

→ Greedy Best First Search

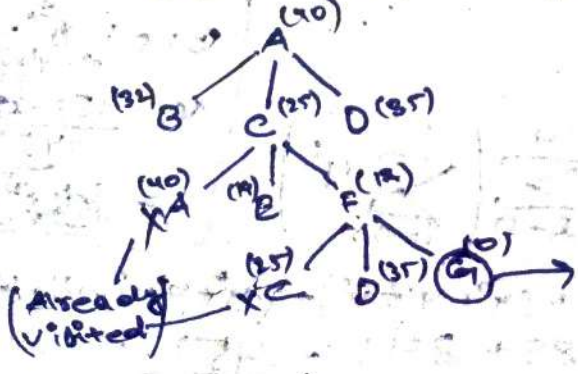
- → $f(n) = h(n)$
- → Incomplete
- → Not optimal search

→ It does n't depend on edge weight



A — 40
B — 32
C — 25
D — 35
E — 19
F — 17
H — 10
G — 0

Heuristic Table

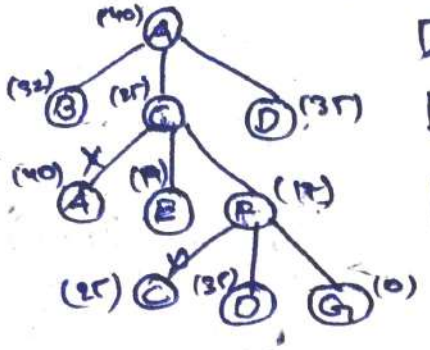


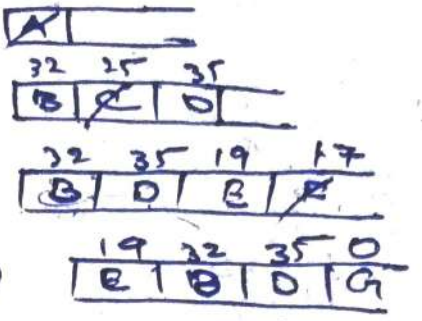
Path: A → C → F → G

Total Path Cost = 44

→ Best First Search
↳ f(n) = h(n)
↳ Priority Queue

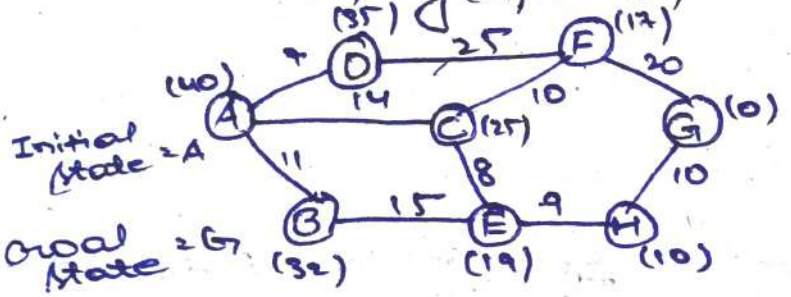
Ex. Same example as previous one

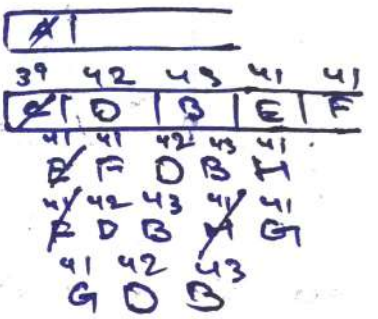


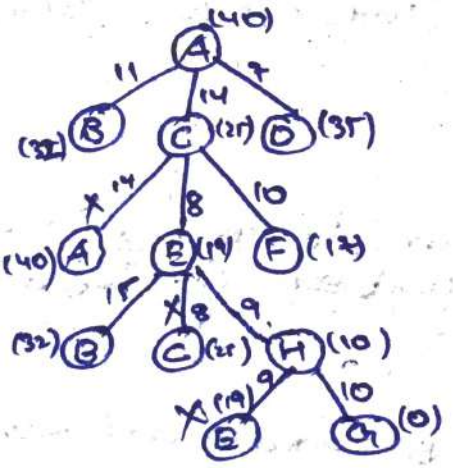




→ A* Search
↳ Priority Queue
↳ f(n) = g(n) + h(n)







$f(B) = g(B) + h(B) = 11 + 32 = 43$

$f(C) = 14 + 25 = 39$

$f(D) = 7 + 35 = 42$

$f(E) = 14 + 8 + 19 = 41$

$f(F) = 14 + 10 + 17 = 41$

$f(H) = 14 + 8 + 9 + 10 = 41$

$f(B) = 14 + 8 + 15 + 32 = 69$

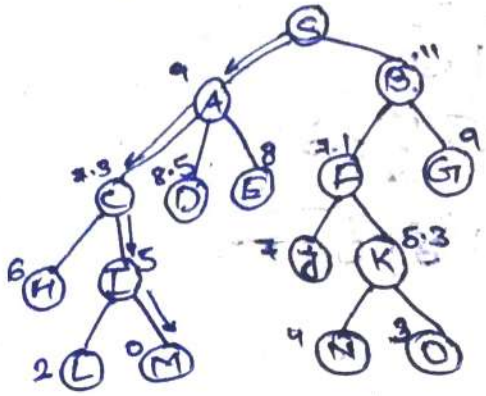
$f(G) = 14 + 8 + 9 + 10 + 0 = 41$

- If heuristically admissible then this search is optimal search.
- Complete search.

Learn AO* search

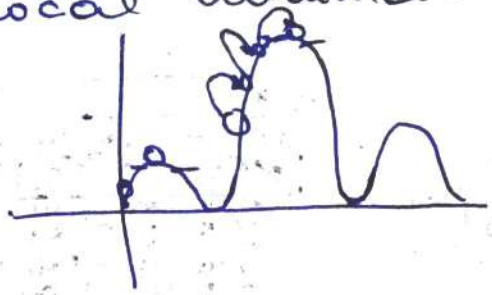
→ Local search
↳ Hill climbing

- less memory.
- move only to neighbour of that node



$$LOSS = \frac{(4-2)^2}{2}$$

- Limitation: It gets stuck on local maxima or local minima.



- Plateau: A flat area of search space in which all neighbouring states have same values.
- Ridge

Ex:

Initial

2	8	3
1	6	4
7		5

Goal

1	2	3
8		4
7	6	5

- heuristic value is the number of misplaced values

Soln:

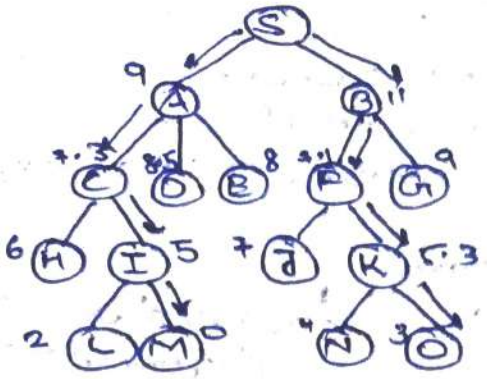
2	8	3
1		4
7	6	5

h=3

(We have to draw all possible options)

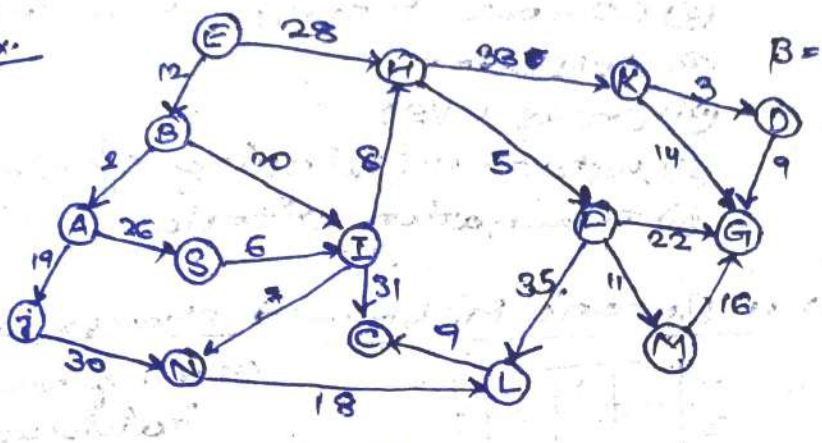
→ Beam search
↳ Select k best search.

β → Beam value



(Select 2 best nodes)

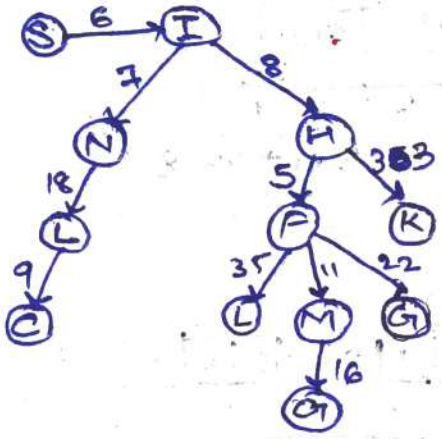
Ex.



B = 2

Go from S to G

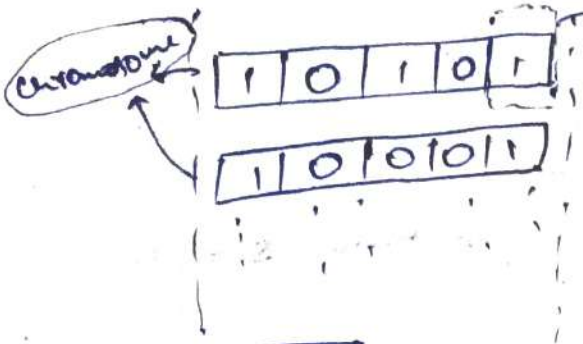
Soln.



∴ Min. cost = 46.

→ Genetic Algorithm
↳ Inspired by Charles Darwin.
↳ Stochastic Beam Search
↳ Randomly k successors to explore

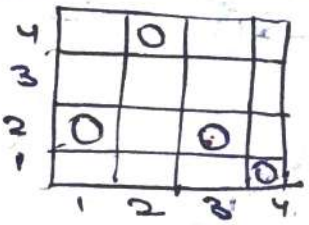
① Initialize population → collection of chromosomes



② Compute Fitness value (map f)
③ Selection
④ Cross over
⑤ Mutation (flipping of bits)
⑥ Termination criteria

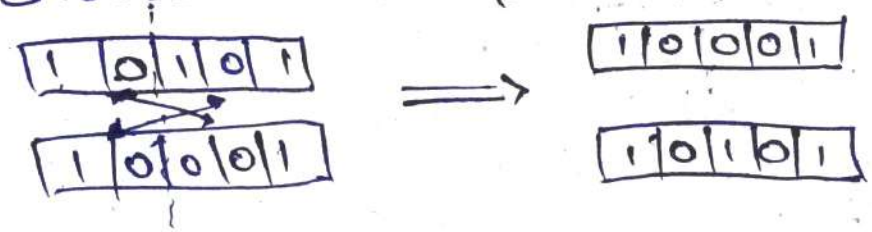
e.g. 2 4 4 1 = 8 (1 2 3 4) f.v
3 1 2 4 =
2 1 1 4 =
3 1 4 1 =

4 Queen's Problem ⇒ Fitness value
f: No. of non-attacking individuals



f = 1+3+1+2 = 8

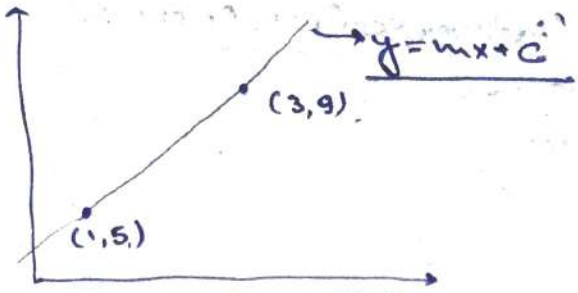
eg: Crossover example



* GA Algo.

① Generate a random population of chromosomes (this is the goal).
② If termination criteria are satisfied, stop. Otherwise, continue with step 3.
③ Determine the fitness of each chromosome
④ Apply crossover/mutation to selected chromosomes from the current generation to generate a new population of chromosomes the next generation.
⑤ Return to step-2.

Ex-



m, c = ?

$$L = \sum (y_{actual} - y_{pred})^2$$

we have to minimize loss

① [m, c] = { (2, 7), (1, 3) }

for [m, c] = (2, 7) and (1, 3),

$y_1 = 2.x + 7$

(1,5) $y_1 = 2.1 + 7 = 9$

(3,9) $y_1 = 2.3 + 7 = 13$

$L = (5-9)^2 + (9-13)^2 = 32$

$y_2 = 1.x + 3$

(1,5) $y_2 = 1.1 + 3 = 4$

(3,9) $y_2 = 1.3 + 3 = 6$

$L = (4-5)^2 + (9-6)^2 = 10$

- Now, we will do mutation on (1,3) to (3,3)
  (001 011 → 011 011)

  So, $y_3 = 3.x + 3$

  (1,5) $y_3 = 3.1 + 3 = 6$

  (3,9) $y_3 = 3.3 + 3 = 12$

  $L = (6-5)^2 + (12-9)^2 = 10$

- mutation on (3,3) to (2,3) —

  So, $y_4 = 2.x + 3$

  (1,5) $y_4 = 2.1 + 3 = 5$

  (3,9) $y_4 = 2.3 + 3 = 9$

  $L = (5-5)^2 + (9-9)^2 = 0$

So, we have got the required line

∴ $y = 2x + 3$ is the fittest line.

→ Gaming Algorithm (Adversarial Search)

↳ Multi-agent Systems

↳ Competitive env.

① Minimax algorithm

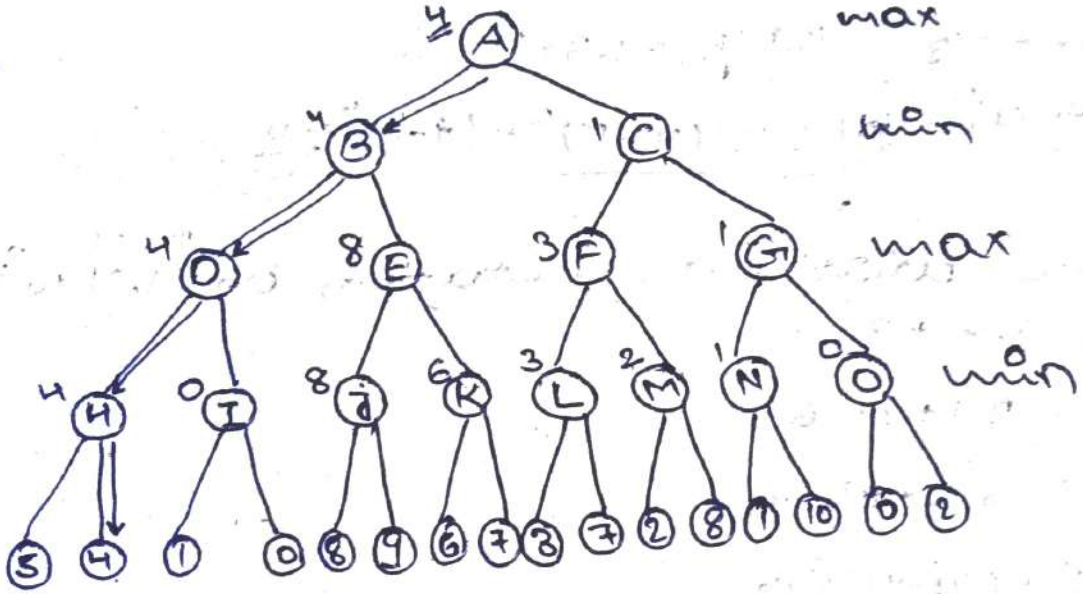
- tic-tac-toe



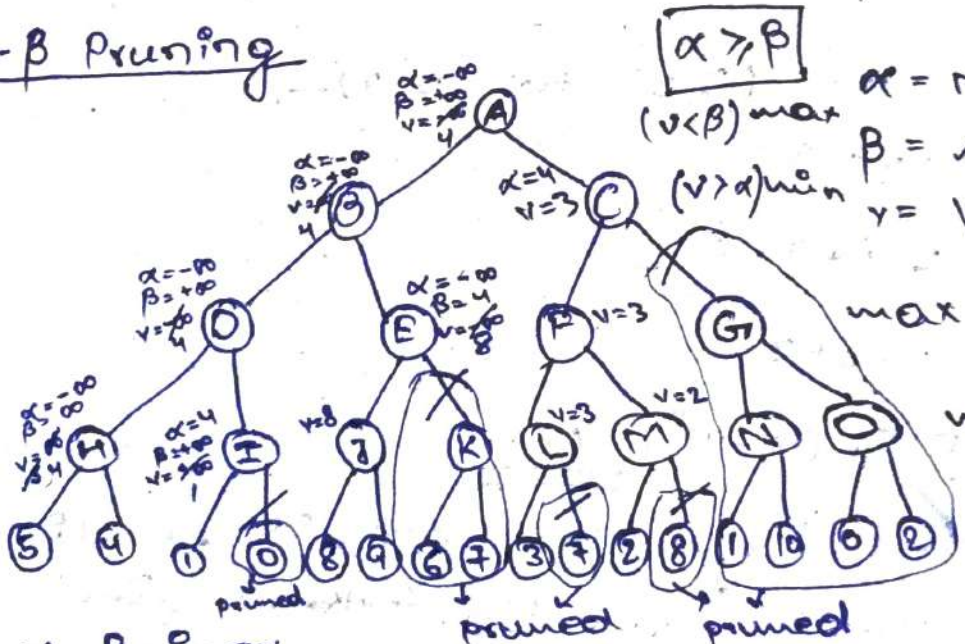
: 9 ways

Total : 9! ways

8 ways

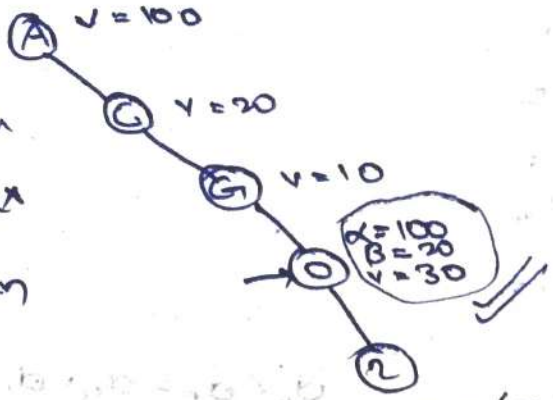


② α-β Pruning

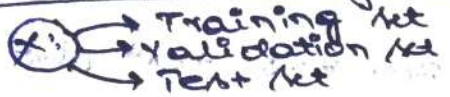


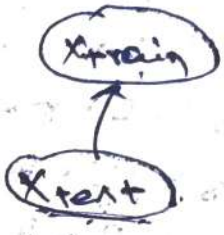
- α = Maximizer
  ↳ Max. value at max level nodes from current to root.
- β = Minimizer
  ↳ Min. value at min level nodes from current to root
- v = value of node

** e.g., max / min / max / min



→ K-Nearest Neighbors (K-NN)





- Supervised learning
  Unsupervised Learning
  Reinforcement learning
  ↓
  Rewards / Penalty

Ex

5	6	8	+	d_1
2	3	5	+	d_2
2	1	5	−	d_3
1	5	6	−	d_4
2	1	3	+	d_5
6	2	8	?	

test →

Euclidean Dist.

$d_5 < d_3 < d_2 < d_1 < d_4$

+ − + + −

Let K = 3, 2 are +ive and 1 -ive so, test data should be +ive.

- $L_p\text{-norm} = \{\sum (x_i - x_j)^p\}^{1/p}$

  ↓

  $L_2\text{-norm} = \{(x_1 - x_2)^2 + (y_1 - y_2)^2\}^{1/2}$

  ↓ Euclidean Distance

- Manhattan Distance L_1-norm:

  $L_1 = |x_1 - x_2| + |y_1 - y_2|$

- Minkowski Distance:

  $L_p\text{-norm} = \{(x_1 - x_2)^p + (y_1 - y_2)^p\}^{1/p}$

Q.

5	6	8	→	d_1
2	3	5	→	d_2
2	1	5	−	d_3
1	5	6	−	d_4
2	1	3	→	d_5
test → 4	8	1	?	

Apply L_1-norm:

$d_1 = 1 + 2 + 7 = 10$

$d_2 = 2 + 5 + 4 = 11$

$d_3 = 2 + 7 + 4 = 13$

$d_4 = 3 + 3 + 5 = 11$

$d_5 = 2 + 7 + 2 = 11$

$d_1 < d_2 = d_4 = d_5 < d_3$

\+ \+ − \+ −

For k=3, test should be +ive.

For tie, take mind

→ Bayesian ~~theorem~~ Algorithm (Naive Bayes)

- Bayes' Theorem:

$$P(X/Y)\ P(Y/X) = \frac{P(X/Y) \cdot P(Y)}{P(X)}$$

$P(Y/X)$: Posterior; $P(X/Y)$: Likelihood; $P(Y)$: Prior; $P(X)$: Evidence

Ex.

f_1	f_2	f_3	y
−	−	−	

$P(Y) = 9/14$, $P(N) = 5/14$

$$P(y=yes/X=\{f_1,f_2,f_3\}) = \frac{P(X=\{f_1,f_2,f_3\}/y=yes) \cdot P(y=yes)}{P(X)}$$

$$= \frac{P(f_1/Y=yes) \cdot P(f_2/Y=yes) \cdot P(f_3/y=yes) \cdot P(y=yes)}{P(X)}$$

$$P(Y=No/X=\{f_1,f_2,f_3\}) = \frac{P(X=\{f_1,f_2,f_3\}/Y=No) \cdot P(y=No)}{P(X)}$$

$$= \frac{P(f_1/No) \cdot P(f_2/No) \cdot P(f_3/No) \cdot P(No)}{P(X)}$$

Ex.

Day	Outlook	Temp	Humidity	Wind	play Tennis
1	Sunny	Hot	High	Weak	NO
2	Sunny	Hot	High	Strong	NO
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	NO
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	NO
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	NO

Test → Overcast Hot High Weak ?

So, $P(Yes) = 9/14$, $P(NO) = 5/14$

Outlook	Yes	NO
Sunny	2	3
Overcast	4	0
Rain	3	2

Temp	Yes	NO
Hot	2	2
Mild	4	2
Cool	3	1

Humidity	Yes	NO
High	3	4
Normal	6	1

Wind	Yes	NO
Strong	3	3
Weak	6	2

Now, $P\left(Y \middle/ X=\{O,H,H,W\}\right) = \dfrac{P(O/Y)\cdot P(H/Y)\cdot P(High/Yes)\cdot P(W/Y)\cdot P(Yes)}{P(X)}$

$= \dfrac{4/9 \cdot 2/9 \cdot 3/9 \cdot 6/9 \cdot 9/14}{P(X)}$

And, $P\left(N \middle/ X=\{O,H,H,W\}\right) = \dfrac{0/5 \cdot 2/5 \cdot 4/5 \cdot 2/5 \cdot 5/14}{P(X)} = 0$

$\because P(Y/X) > P(N/X)$

$\therefore$ Tennis will be played on such a day

* Laplace Smoothing
  → To remove the probability of 0, we use this.

- $$P_{Lap}(X/Y) = \frac{C(x,y)+\alpha}{N+\alpha \cdot |X|}$$ → (No. of diff variables)

No, $P_{Lap}\left(\frac{\text{Overcast}}{\text{No}}\right) = \frac{C(0,N)+\alpha \cdot 1}{5+\alpha \cdot 3} = \frac{0+1}{5+3} = \frac{1}{8}$

Also, $P_{Lap}\left(\frac{\text{Mild}}{\text{Yes}}\right) = \frac{C(M,Y)+\alpha}{N+\alpha|X|} = \frac{4+1}{9+1 \cdot 3} = \frac{5}{12}$

* For continuous datasets like temp. degrees, we will use gaussian distn.:

$$P = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{\frac{-(x-\mu)^2}{2\sigma^2}} \quad ; \quad \mu = \frac{\Sigma \text{temp}}{N}$$

$$\sigma^2 = \frac{\Sigma(x-\mu)^2}{(N-1)}$$ (Degree of freedom)

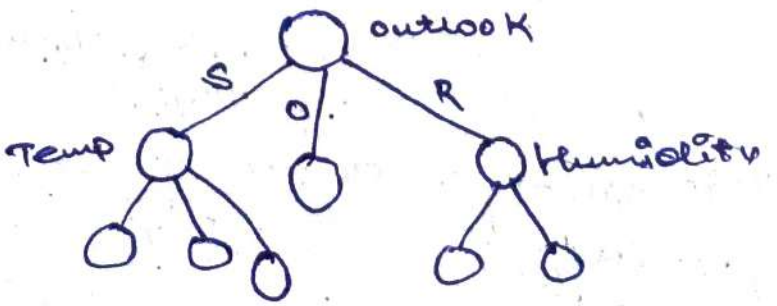
eg: $P\left(\frac{\text{Yes}}{25}\right) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{\frac{-(25-\mu)^2}{2\sigma^2}}$

→ Decision Tree
  → supervised.
  → Non-parametric

e.g:

Outlook		Temp	
Sunny	~~2+~~ 3+, ~~3-~~ 2-	Hot	2+, 2-
Overcast	4+, 0-	Mild	4+, 2-
Rainy	~~2+~~ 3+, ~~3-~~ 2-	Cold	3+, 1-

Humidity		Windy	
High	3+, 4-	False	6+, 2-
Normal	6+, 1-	True	3+, 3-



- Information Gain = Entropy (Parent) − [Average entropy of children]

$$IG(S, X) = Entropy(S) - \sum_{v \in \{v_1, v_2, v_3\}} \frac{|S_v|}{|S|} Entropy(S_v)$$

$$IG(S, Outlook) = Entropy(S) - \sum_{v \in \{Sunny, Overcast, Rainy\}} \frac{|S_v|}{|S|} E(S_v)$$

NOW, $\boxed{Entropy = -\sum P_i \log_2 P_i}$

$$= -\{P_+ \log_2 P_+ + P_- \log_2 P_-\}$$

SO, $E(S) = -\frac{9}{14}\log\frac{9}{14} - \frac{5}{14}\log\frac{5}{14}$

$E(S_{Sunny}) = -\frac{2}{5}\log\frac{2}{5} - \frac{3}{5}\log\frac{3}{5}$

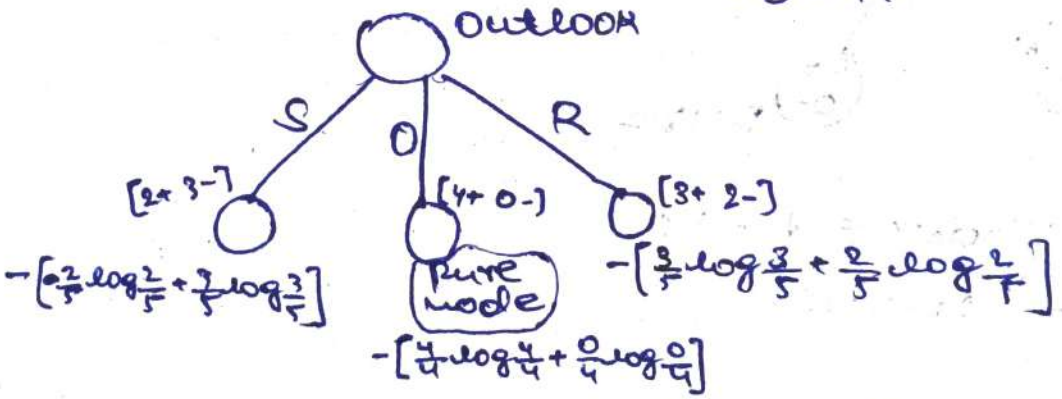
$E(S_{Overcast}) = -\frac{4}{4}\log\frac{4}{4} - \frac{0}{4}\log\frac{0}{4}$

$E(S_{Rainy}) = -\frac{3}{5}\log\frac{3}{5} - \frac{2}{5}\log\frac{2}{5}$

Now, $IG(S, Outlook) = a - \frac{|S_s|}{|S|} * E(S_s) - \frac{|S_o|}{|S|} * E(S_o) - \frac{|S_r|}{|S|} * E(S_r)$

$$= a - \left[\frac{5}{14} * b + \frac{4}{14} * c + \frac{5}{14} * d\right]$$

$= 0.247$

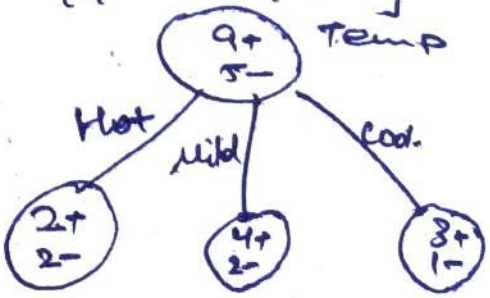


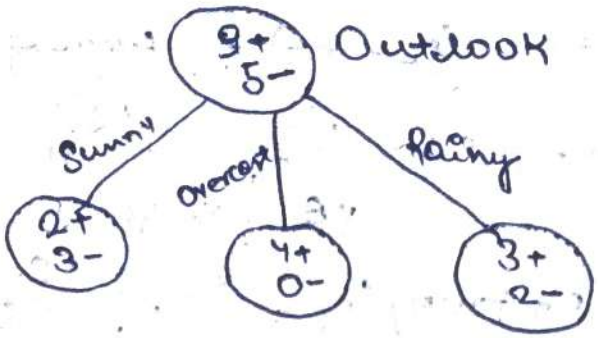
Also, $IG(S, Temp) = p - \left[\frac{4}{14} \times q + \frac{6}{14} \times r + \frac{4}{14} \times s\right]$

$= 0.029$

$IG(S, Humidity) = 0.152$

$IG(S, Windy) = 0.048$





Temp

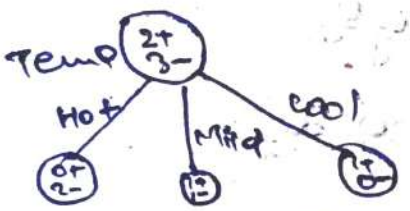
$S_{sunny}[2+, 3-] \rightarrow -\left[\frac{2}{5}\log\frac{2}{5} + \frac{3}{5}\log\frac{3}{5}\right]$

$S_{Hot}[0+, 2-] \rightarrow -\left[\frac{0}{2}\log\frac{0}{2} + \frac{2}{2}\log\frac{2}{2}\right]$

$S_{Mild}[1+, 1-] \rightarrow -\left[\frac{1}{2}\log\frac{1}{2} + \frac{1}{2}\log\frac{1}{2}\right]$

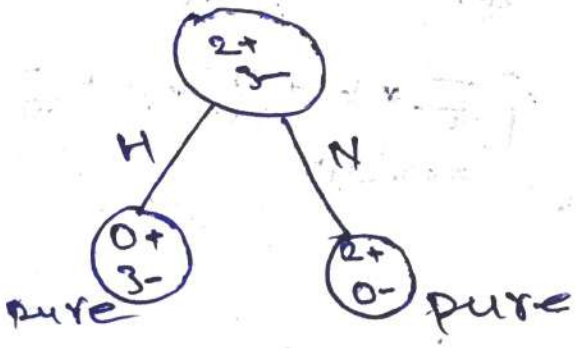
$S_{Cool}[1+, 0-] \rightarrow -\left[\frac{1}{1}\log\frac{1}{1} + \frac{0}{1}\log\frac{0}{1}\right]$

$$IG(S_{sunny}, Temp) = E(S_{sunny}) - \sum_{v \in \{H, M, C\}} \frac{|S_v|}{|S_{sunny}|} \cdot E(S_v)$$



$$= E(S_{sunny}) - \left[\frac{2}{5} \times E(S_{Hot}) + \frac{2}{5} \times E(S_{Mild}) + \frac{1}{5} \times E(S_{Cool})\right]$$

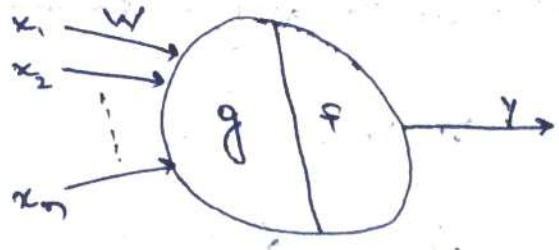
Humidity



- Gini Index = $\{1 - P_i\}^2$

  Chi-Square =

Neuron



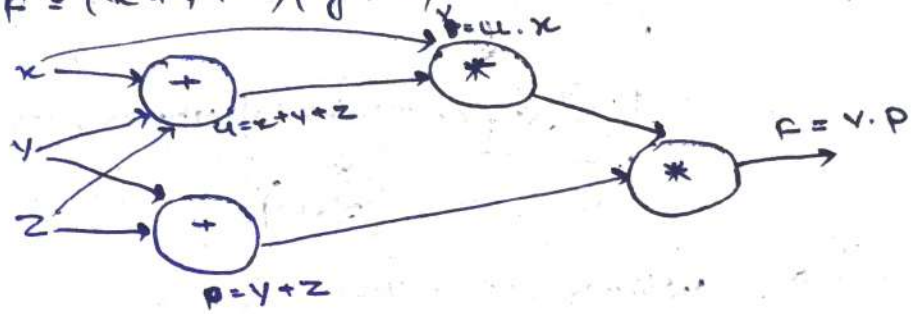
$g = \sum x_i = x_1 + x_2 + \dots + x_n$

$y = f(g)$

$y = \begin{cases} 1 & \text{if } f(g(x)) \geq \theta \\ 0 & \text{otherwise } y = 0 \end{cases}$

e.g. Computation Graph → (practice it!)

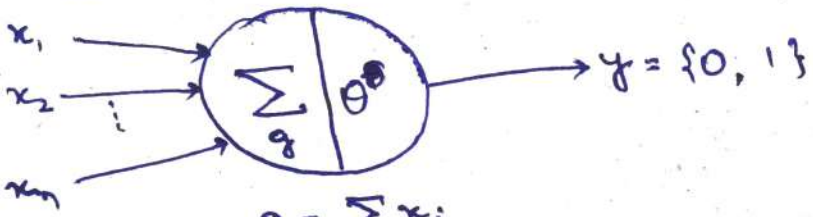
$f = (x+y+z)(y+z)x$



* MP Neuron

↓

"McCulloch-Pitts neuron"



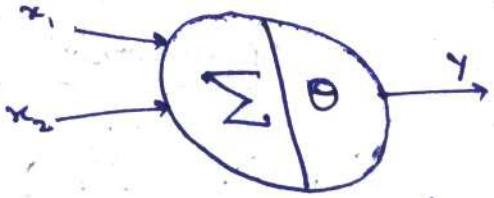
$g = \sum x_i$

$x \in \{0, 1\}$ if $\sum x_i \geq \theta \Rightarrow y = 1$

otherwise $\Rightarrow y = 0$

e.g. OR

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1



$\sum x_i \geq \theta \Rightarrow y = 1$

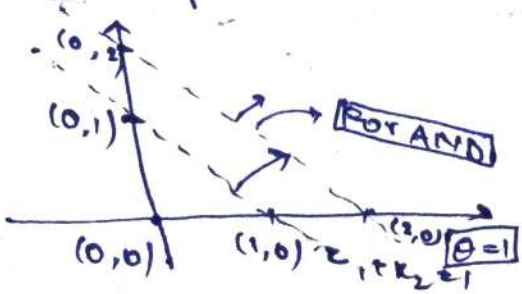
$x_1 + x_2 \geq \theta ; \quad y = 1$

$1 + 1 \geq \theta ; \quad y = 1$

$1 + 0 \geq \theta ; \quad y = 1$

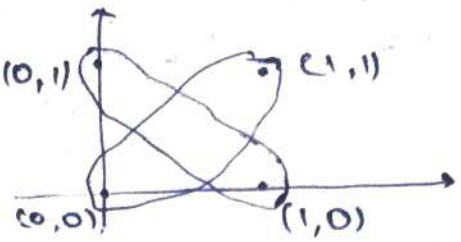
$0 + 1 \geq \theta ; \quad y = 1$

$0 + 0 \quad ; \quad y = 0$



eg: XOR → Not possible with MP-Neuron
→ Requires multilayer perceptron

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0



✓ not linearly separable

- $\sum w_i x_i \geq \theta$

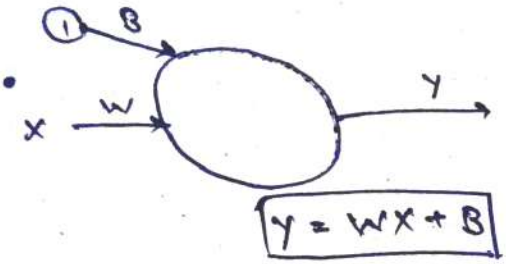
$w_1 x_1 + w_2 x_2 \geq \theta$

$w_1 x_1 + w_2 x_2 - \theta \geq 0$

$\sum w_i x_i - \theta \geq 0 \Rightarrow$ $\boxed{\sum w_i x_i - \theta = 0}$

$\boxed{\sum_i w_i x_i + w_0 = 0}$ where, $w_0 = -\theta$

(bias)

- 

$y = wx + B$

$wx + B \geq 0 \Rightarrow y = 1$

Otherwise $\Rightarrow y = 0$

- $L = \sum (y - \hat{y})^2$

$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$

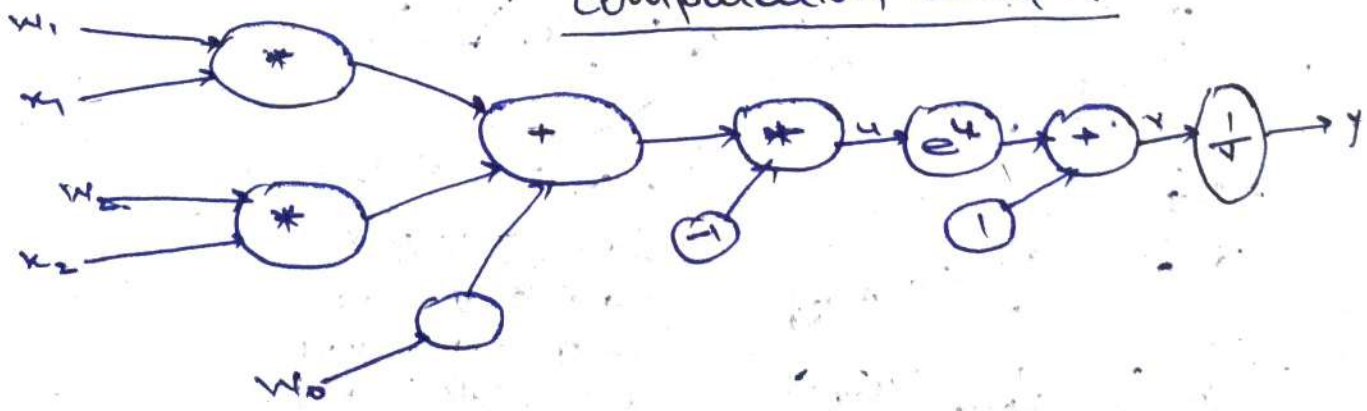
$B_{new} = B_{old} - \alpha \frac{\partial L}{\partial B}$

↓

learning parameter

e.g. $y = \frac{1}{1 + e^{-(w_1 x_1 + w_2 x_2 + w_0)}}$

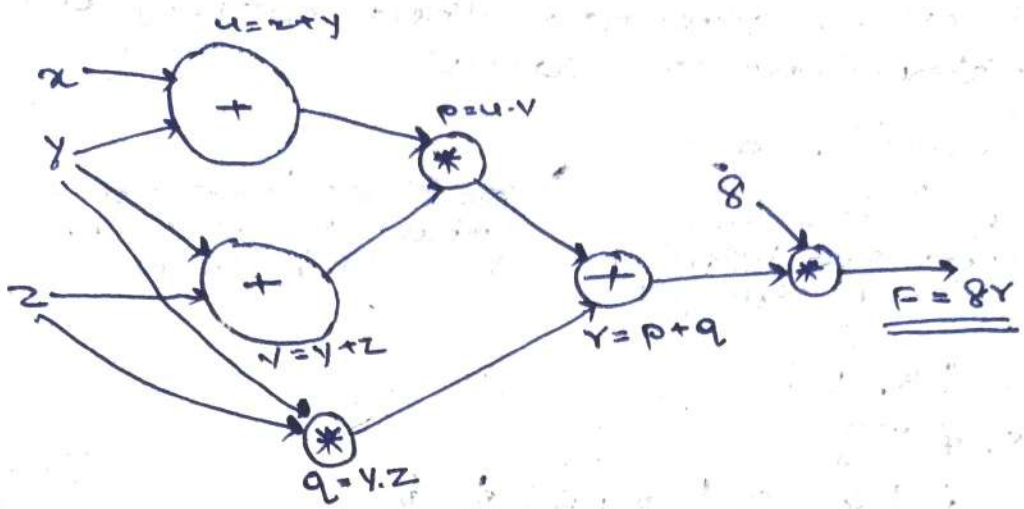
computation Graph



e.g. $f = 8[(x+y)(y+z)+yz]$

Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial z}$.

Soln:

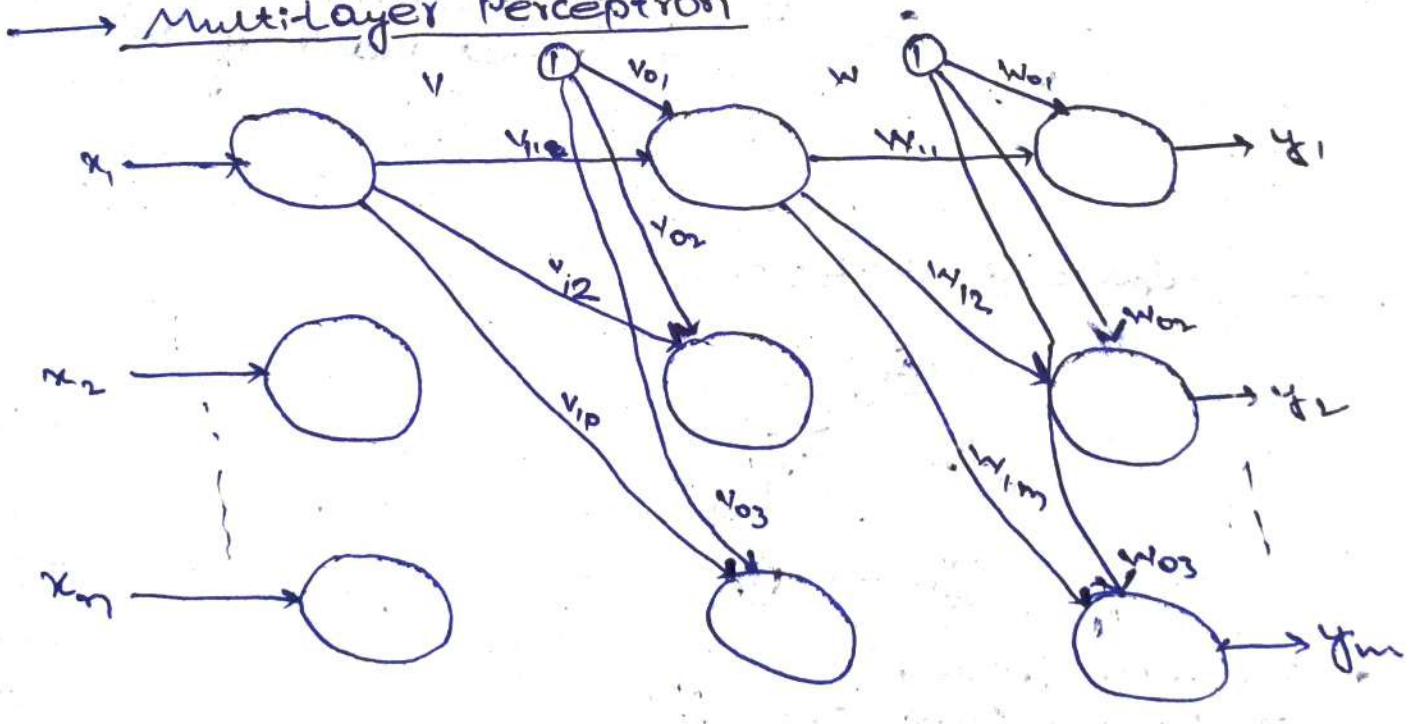


Now, $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial p} \cdot \frac{\partial p}{\partial u} \cdot \frac{\partial u}{\partial x}$

$$\frac{\partial f}{\partial y} = \left(\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial p} \cdot \frac{\partial p}{\partial u} \cdot \frac{\partial u}{\partial y}\right) + \left(\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial p} \cdot \frac{\partial p}{\partial v} \cdot \frac{\partial v}{\partial y}\right) + \left(\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial q} \cdot \frac{\partial q}{\partial y}\right)$$

$$\frac{\partial f}{\partial z} = \left(\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial q} \cdot \frac{\partial q}{\partial z}\right) + \left(\frac{\partial f}{\partial r} \cdot \frac{\partial r}{\partial p} \cdot \frac{\partial p}{\partial v} \cdot \frac{\partial v}{\partial z}\right)$$

→ Multi-Layer Perceptron



i/p layer Middle layer o/p layer

→ Constraint Satisfaction Problem (CSP)

X : Set of variables $\{X_1, X_2, \ldots, X_n\}$

D : Set of Domains $\{D_1, D_2, \ldots, D_n\}$

C : Set of constraints that satisfy allowable combinations of value

C_i = <scope, val>

- scope → (tuple of variable that participate in constraints)
- val → (defines the values that those variables can take)

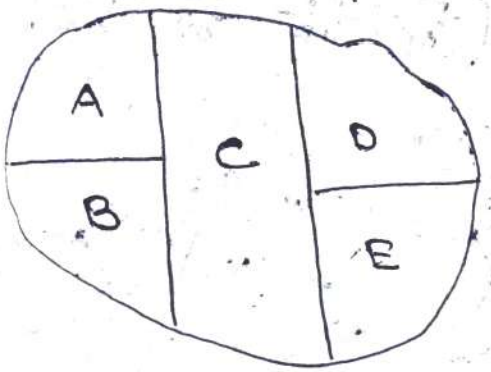
e.g. $X_1, X_2 = \{A, B\}$

$C_i [X_1 \neq X_2]$

$X_1 = A$	$X_1 = B$	~~$X_1 = A$~~	~~$X_1 = B$~~
$X_2 = B$	$X_2 = A$	~~$X_2 = A$~~	~~$X_2 = B$~~

Both can't be explored

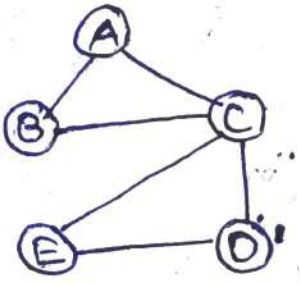
e.g. Graph Coloring Problem



{R, B, G}

Constraints : No adjacent cities having same color.

Soln: Adjacent Graph



- Sudoku Solver
  N-Queen Problem

Cryptoarithmetic Problem

```
  SOME
+ MORE
-------
 MONEY
```

$\{ A \rightarrow \{0-9\}, \ldots, Z \rightarrow \{0-9\} \}$

- $(HE)^H = SHE$
  Find $S + H + E$?

① Each alphabet take only one number from 0 to 9

② Carry over can be –
max 1 when we add 2 numbers
max 2 " " " 3 "
" 3 " " " 4 "
⋮

③ Sum of 2 same nos. must be an even no.

④ If two numbers are A & B, sum of A & B in A then,
- The value of B must be 0 if A & B are unit digits.
- The value of B may be 0 or 9 if A & B are other than unit digits.

⑤ Addition of 2 nos. with n digits, if the result is n+1 digits the left most digit must be 1.

e.g.
```
  SEND
+ MORE
-------
 MONEY
```
→
```
 8 9 END
 10 RE
-------
 10NEY
```
S = {8, 9}

```
 8 END
 10RE
-------
 10NEY
```
→ X

E = {9}

```
 9 END
 10RE
-------
 10NEY
```
(Make combinations and solve)

e.g. $(HE)^H = SHE$

$(25)^2 = 625$

$\therefore S + H + E = 6 + 2 + 5 = \underline{\underline{13}}$.

e.g.
$$\begin{array}{r} T\,\emptyset^{1} \\ +\,G\,\emptyset_{1} \\ \hline 1\,\emptyset\,U\,T \end{array} \longrightarrow \begin{array}{r} {}^{2}\not{8}\,1 \\ +\,G\,1 \\ \hline 1\,U\,\not{T}_{2} \end{array} \longrightarrow \begin{array}{r} 2\;1 \\ +\,{}^{8}\not{G}\;1 \\ \hline 1\,\not{\emptyset}_{0}\,2 \end{array} \longrightarrow \begin{array}{r} 21 \\ +\,8\,1 \\ \hline 102 \end{array}$$

$G = \{8, \not{8}\}$

e.g.
$$\begin{array}{r} Y\;O\;U\;R \\ +\;\;Y\;O\;U \\ \hline H\,E\,A\,R\,T \end{array}$$
(Assume $O = 4$)
Find $Y + U + R + E$

Soln.
$$\begin{array}{r} 9\not{Y}^{+1}\leftarrow\not{\emptyset}^{4}\,U\,R \\ +\;\;9\not{Y}\;\not{\emptyset}_{4}\,U \\ \hline 1\not{H}\not{E}_{0}\,A\,R\,T \end{array} \longrightarrow \begin{array}{r} 9\;4\;\not{\emptyset}^{2}\not{R}^{6} \\ +\;\;9\;4\;\not{U}_{2} \\ \hline 1\;0\;\not{A}_{3}\not{R}_{6}\not{T}_{8} \end{array} \longrightarrow \begin{array}{r} 9426 \\ +\;\;942 \\ \hline 10368 \end{array}$$

~~$A = \{3\}$~~ $U = \{2, 5\}$

$\therefore Y + U + R + E = 9 + 2 + 6 + 0 = \underline{\underline{17}}$